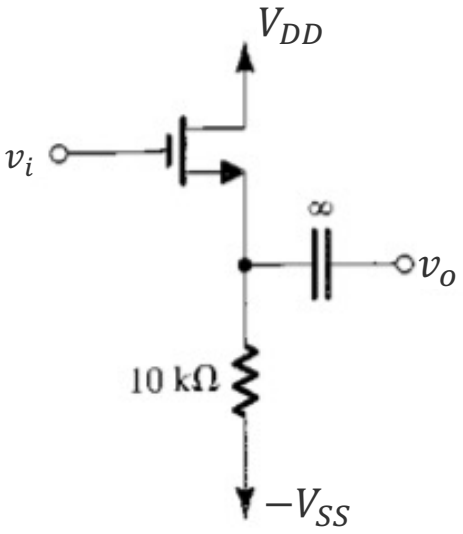
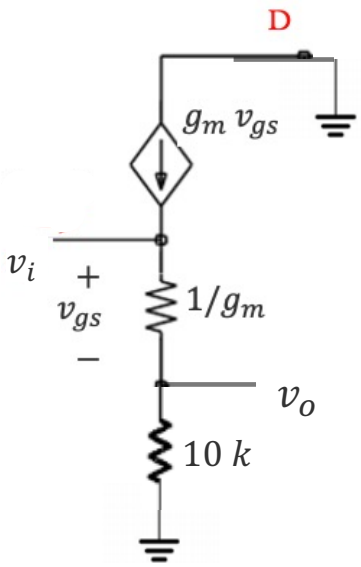
Field Effect Transistors

- MOSFET Basic Amplifier Configurations: Examples

Examples

1) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = v_{gs} + v_o$$

$$v_o = g_m v_{gs}\,(10\ k)$$

$$v_o = g_m(v_i - v_o)(10\ k)$$

$$A_v = \frac{v_o}{v_i} = \frac{g_m\,(10\ k)}{1 + g_m\,(10\ k)}$$

$$A_v = 0.98\ V/V$$

Input Resistance

$$R_i = \infty$$

Output Resistance
short-circuit source v_i

$$R_o = \frac{v_o}{i_o}$$

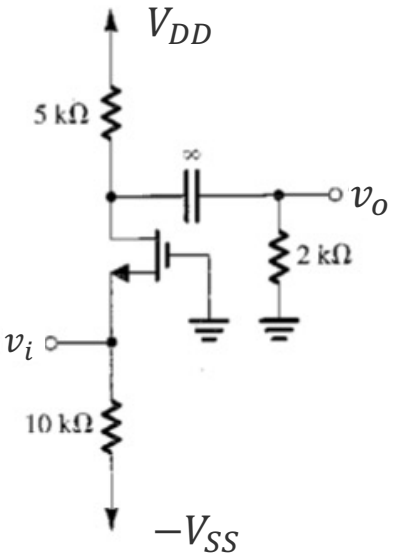
$$i_o = g_m v_{gs} - \frac{v_o}{10\ k}$$

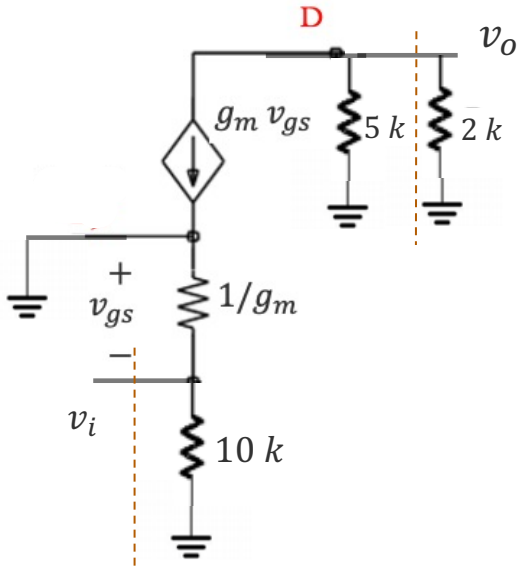
$$i_o = -g_m v_o - \frac{v_o}{10\ k}$$

$$R_o = \frac{1}{g_m + \frac{1}{10k}} = 196.1\ \Omega$$

Examples

2) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = -v_{gs}$$

$$v_o = -g_m v_{gs}\,(5\ k \parallel 2\ k)$$

$$v_o = g_m v_i\,(5\ k \parallel 2\ k)$$

$$A_v = \frac{v_o}{v_i} = g_m(5\ k \parallel 2\ k)$$

$$A_v = 7.14\ V/V$$

Input Resistance

$$R_i = (10\ k \parallel 1/g_m)$$

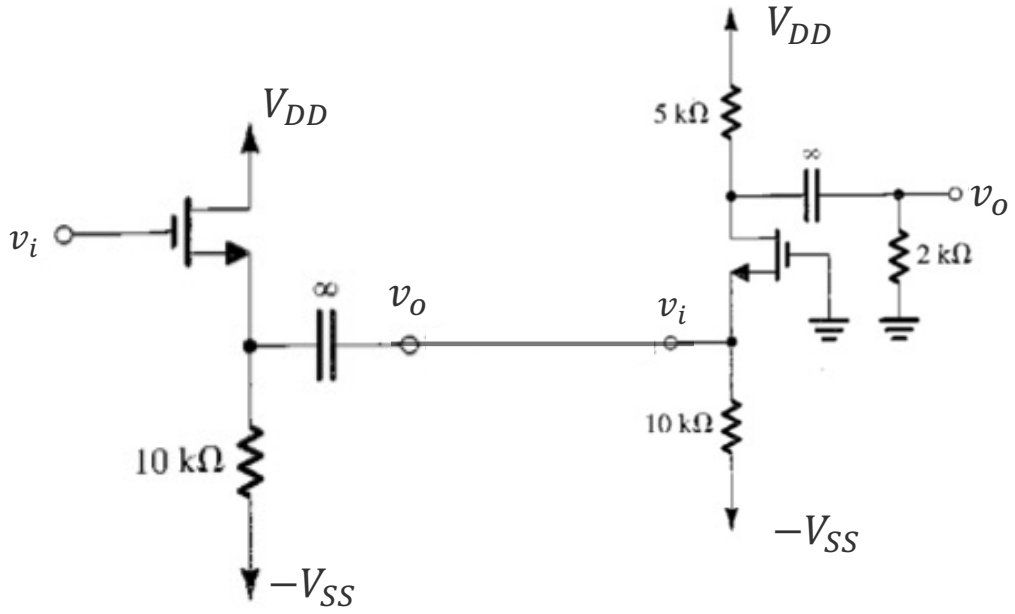
$$R_i = 196.1\ \Omega$$

Output Resistance
short-circuit source v_i

$$R_o = 5\ k\Omega$$

Examples

3) Connect the two circuits in examples 1 and 2.



Multi-stage Amplifier Design

Voltage gain

$$A_v = \left(\frac{v_o}{v_i}\right)_1 \cdot \left(\frac{v_o}{v_i}\right)_2$$

$$A_v = 7\ V/V$$

Input Resistance

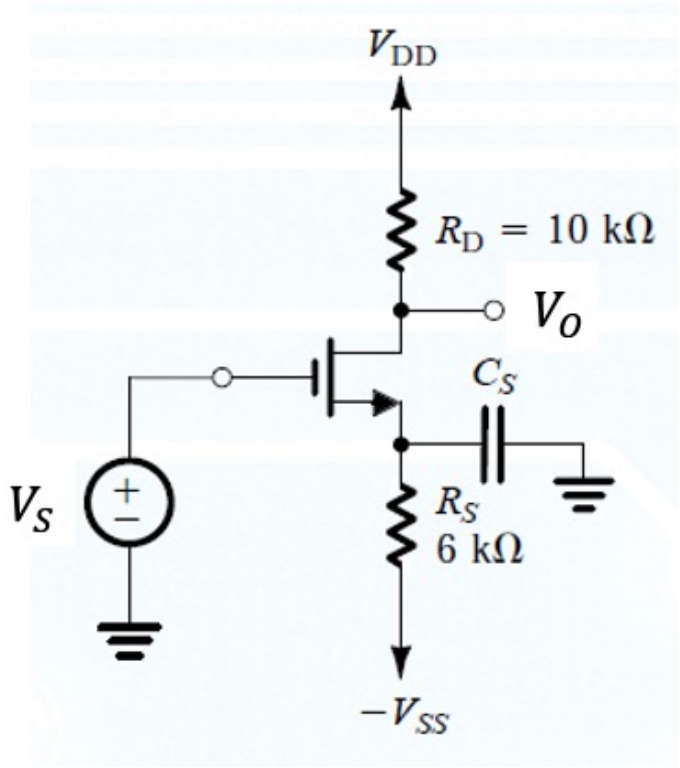
$$R_i = \infty$$

Output Resistance

$$R_o = 5\ k\Omega$$

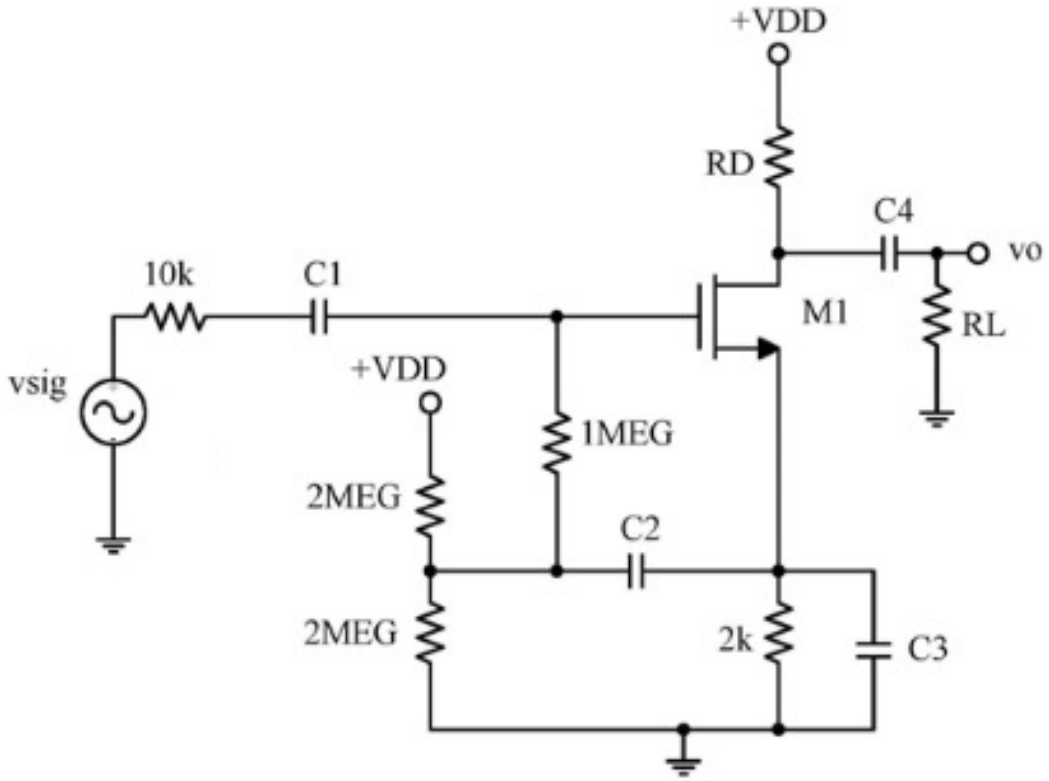
Examples

(4) Analyze the amplifier given below.



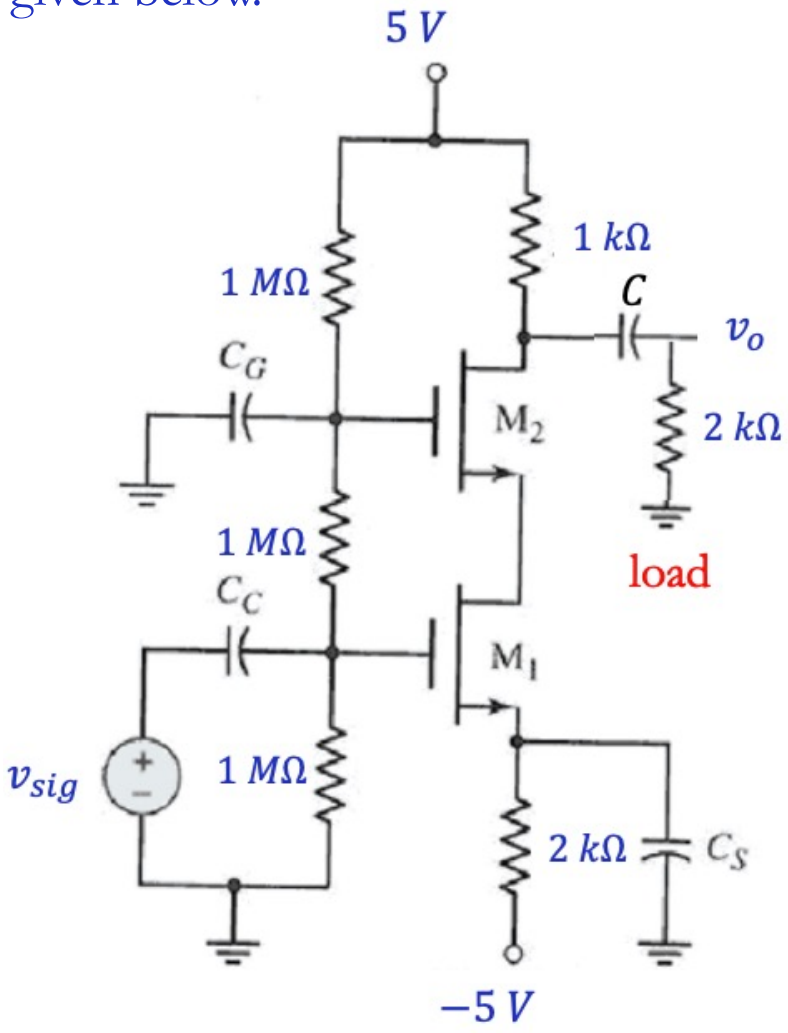
Examples

(5) Analyze the amplifier given below.



Examples

(6) Analyze the amplifier given below.

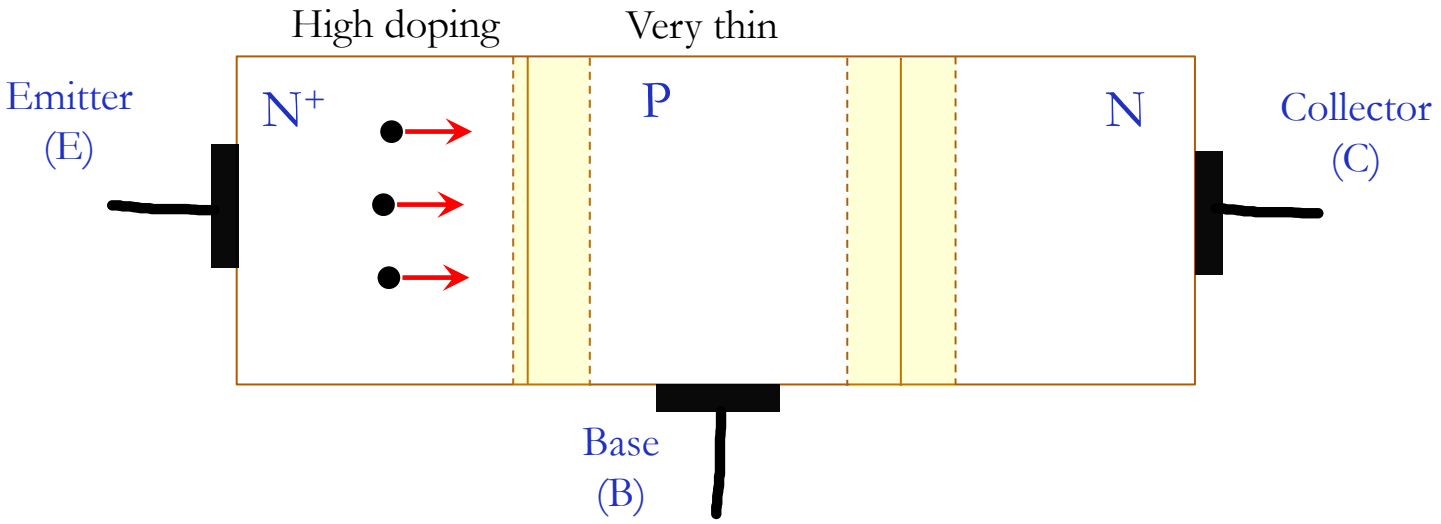


Bipolar Junction Transistors

- BJT Basic Structure.
- Current in BJTs and Operation Modes.
- DC Analysis (Biasing).
- Small Signal Analysis.
- The BJT as an Amplifier.

Device Structure

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.

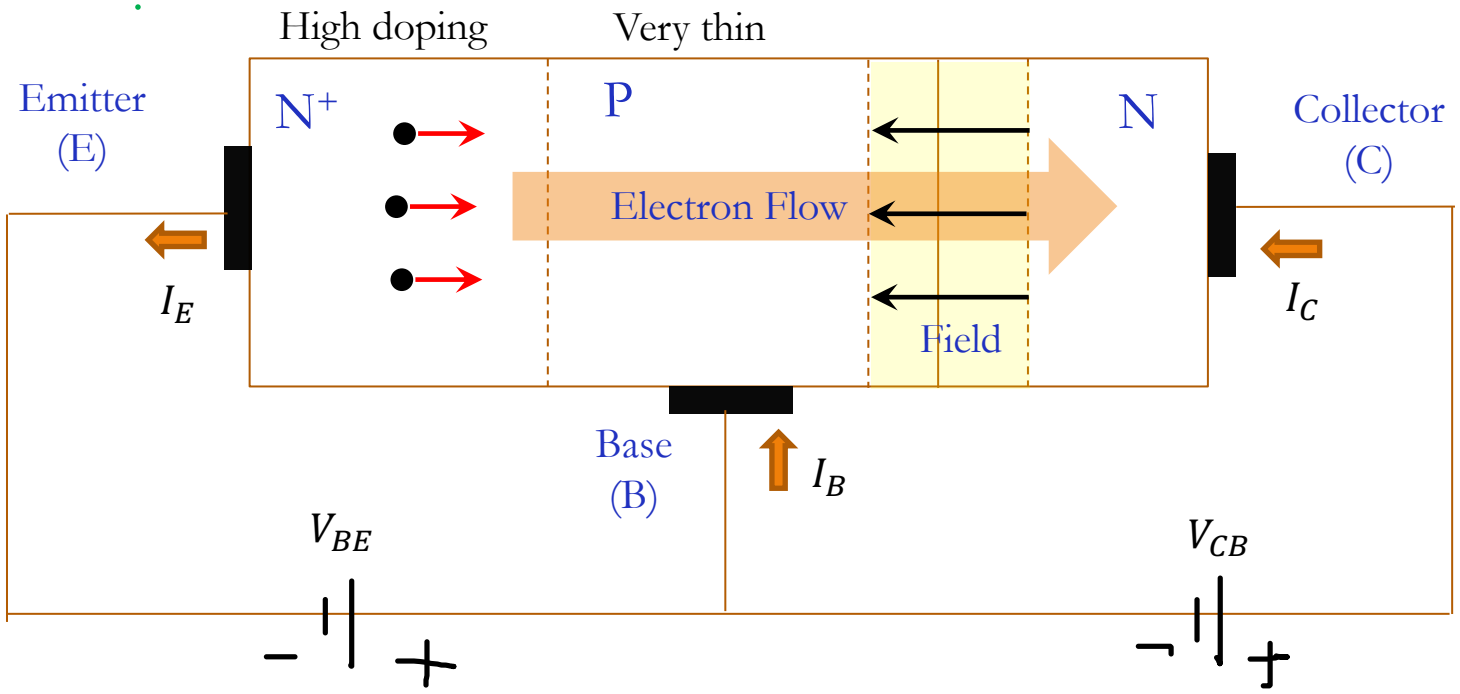


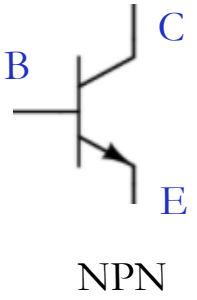
Four possible scenarios for current.

1) The two junction are reversed biased (cutoff mode).
2) BE junction forward, BC junction reversed (active mode).
3) BE junction reversed, BC junction forward (reverse active mode).
4) The two junction are forward biased (saturation mode).

The Active Mode

The active mode is the mode of operation we use in amplifier circuit.





Current relations

$V_{BE} = 0.7\ V$

$I_E = I_B + I_C$

$I_E = (\beta + 1) I_B$

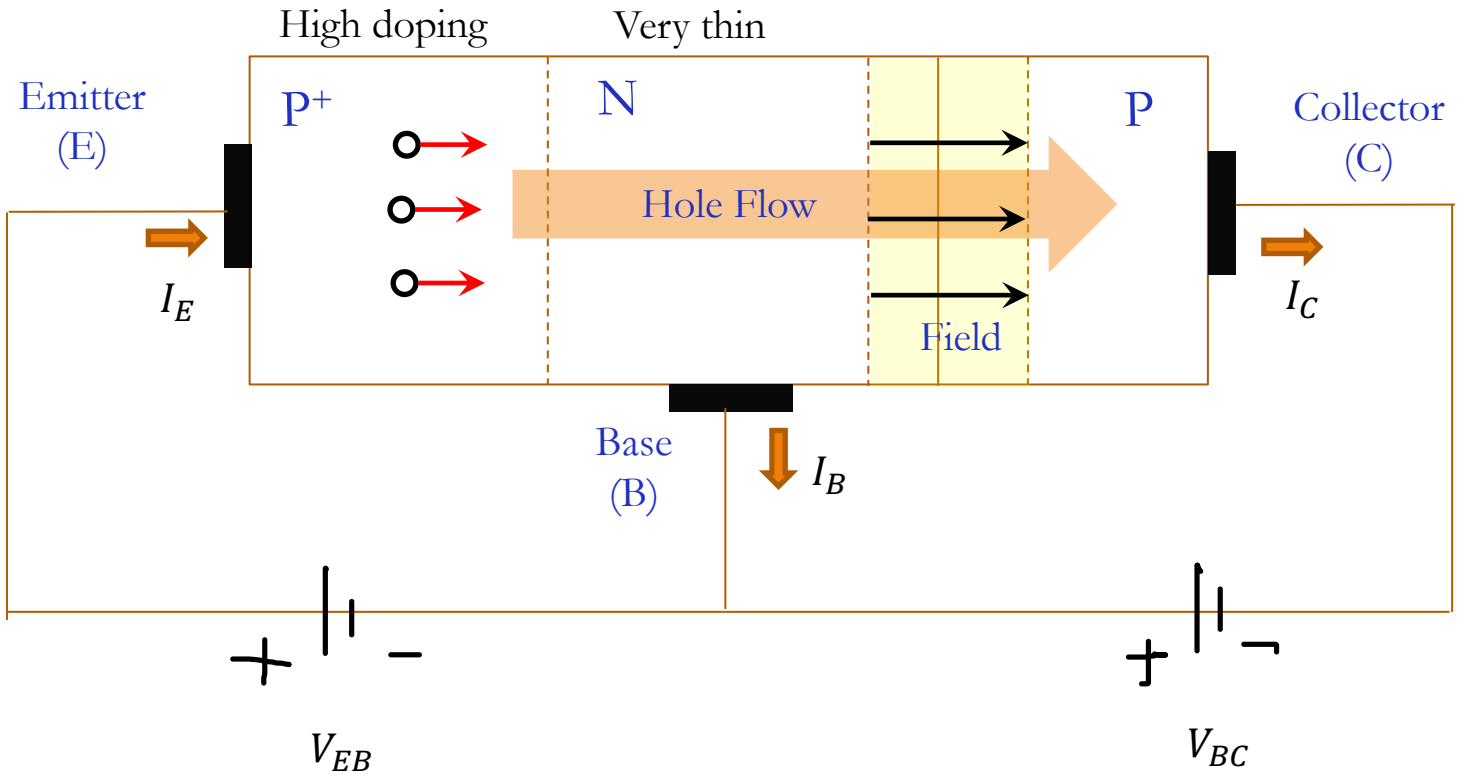
$\beta = \frac{I_C}{I_B}$

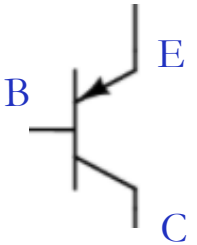
$\alpha = \frac{I_C}{I_E}$

$\alpha = \frac{\beta}{\beta + 1}$

The Active Mode

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.





PNP

Current relations

$$V_{EB} = 0.7\ V$$

$$I_E = I_B + I_C$$

$$I_E = (\beta + 1) I_B$$

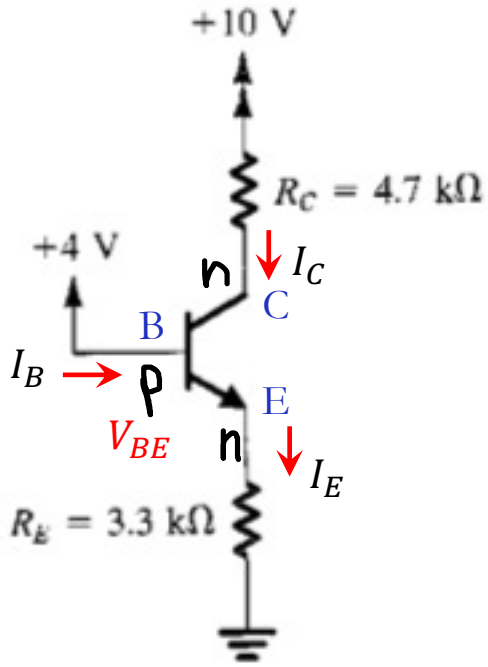
$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

Examples

1) Analyze the DC BJT circuit below. (β=100).



Assume active mode

$$V_{BE} = 0.7\,V = V_B - V_E$$

$$V_B = 4\,V$$

$$V_E = 4 - 0.7 = 3.3\,V$$

$$I_E = 1\,mA$$

$$I_C = \frac{\beta}{\beta+1} I_E = \frac{100}{101} 1\,mA = 0.99\,mA$$

$$I_B = I_E - I_C = 10\,\mu A$$

$$V_C = 10 - I_C R_C = 5.35\,V$$

$$V_C > V_B$$

CB junction is reversed biased

Assumption is correct

Q & A

Send me an email if
you have any
questions.